Name: Arshad Ali
Roll No: 2k23/CSEE/16
Assignment: #1

Assignment 1: AI System Specification

Course: Introduction to Artificial Intelligence

Selected System: Option E — Smart Traffic Signal Controller

Task 1: PEAS Specification

Performance Measures

- **Average Wait Time:** Minimize the idle time for vehicles at red lights.
- **Traffic Throughput:** Maximize the volume of vehicles passing per hour.
- **Emergency Priority:** Rapid detection and clearance for ambulances/fire trucks.
- **Carbon Footprint:** Reduction in emissions through smoother traffic flow.

Environment

Urban city intersections with dynamic variables such as varying traffic density, pedestrians, weather conditions (rain/fog), and unexpected road hazards or accidents.

Actuators

- **Signal Relays:** Switching logic for Red, Yellow, and Green lights.
- **VMS Boards:** Variable Message Signs to provide real-time traffic updates.
- **Audio Warnings:** Signal sounds for visually impaired pedestrians.

Sensors

- **High-Resolution Cameras:** Real-time visual tracking of vehicle counts.
- **Inductive Loops:** Electromagnetic sensors in the asphalt to detect metallic mass.
- **Acoustic Sensors:** Microphones tuned to emergency vehicle siren frequencies.

Task 2: Environment Classification

- **Partially Observable:** The agent only sees the local intersection; distant traffic is unknown.
- **Multi-agent:** Must coordinate with neighboring signals to manage a grid efficiently.

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- **Stochastic:** Unpredictable events like car breakdowns occur randomly.
- **Sequential:** Current timing affects future congestion levels in later cycles.
- **Dynamic:** The environment state changes constantly while the agent is processing.
- **Continuous:** Velocity and time are measured on an infinite real-number scale.

Task 3: Critical Analysis

Greatest Technical Challenge: The *Stochastic* nature of the environment. Managing unpredictable human behavior and sudden road incidents requires complex probabilistic modeling to ensure safety and efficiency.

Utility Function: A proposed function is $U = 0.7 \times Throughput - 0.3 \times Wait_Time$. Doubling the weight of throughput would prioritize traffic volume over individual wait times on minor roads.

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Task 4: Structured Representation (JSON)

```
{
  "system_name": "Smart Traffic Signal Controller",
  "peas": {
    "performance_measures": [
      "average_wait_time",
      "traffic_throughput",
      "emergency_clearance",
      "emission_reduction"
    ],
    "environment": "Dynamic urban intersection grid",
    "actuators": ["signal_lights", "vms_displays", "audio_alerts"],
    "sensors": ["rgb_cameras", "inductive_loops", "acoustic_sensors"]
  },
  "environment_classification": {
    "observability": {
      "choice": "Partially Observable",
      "justification": "Limited sensor range prevents full city awareness."
    },
    "agents": {
      "choice": "Multi-agent",
      "justification": "Requires coordination with adjacent intersection
controllers."
    },
    "predictability": {
      "choice": "Stochastic",
      "justification": "Human behavior and accidents are non-deterministic."
    }
  },
  "utility_function": "U = 0.7*throughput - 0.3*wait_time"
}
```


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